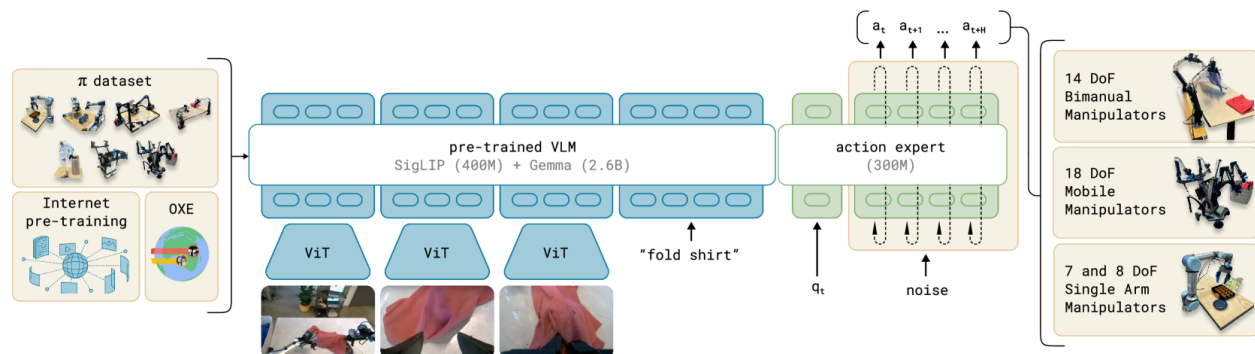


Benchmark of Vision-Language-Action Model for Pharmaceutical Process Interventions



Description: This project will investigate the capabilities of the vision-language-action (VLA) model π_0 when applied to practical use-cases in the pharmaceutical industry. This industry is of special interest, as a lot of focus is placed on avoiding human contamination of products through automation of manual processes; However, when automation fails or unexpected issues occur, a human is still required to step in and manually perform an intervention. VLA models have demonstrated that robot manipulators can perform abstract and complex manipulations based on image and natural language instruction pairs, and therefore, this project will investigate how π_0 can be finetuned to perform specific stochastic interventions to further reduce the risk of contamination.

The ambition is to derive a model that can be used to perform interventions remotely by simply describing what the robot should do at a high level. The project will be carried out at Novo Nordisk's Robotics Innovation Lab in Hillerød, where the student will be using a VR-based teleoperation setup to collect data from a simulated intervention scenario, and use that data to finetune π_0 through imitation learning. To benchmark the model, it is of interest to document how well it handles stochastic conditions and distractors, reflective surfaces, and changes in lighting conditions.

Keywords: vision-language-action models, multimodal learning, vision-based manipulation, generalist robot policies

Resources:

[1] π_0 : A Vision-Language-Action Flow Model for General Robot Control

(<https://arxiv.org/html/2410.24164v1>)

[2] Large VLM-based Vision-Language-Action Models for Robotic Manipulation: A Survey

(<https://arxiv.org/pdf/2508.13073>)

[3] Interfering with Interventions in Aseptic Processing

(<https://www.pharmtech.com/view/interfering-with-interventions-in-aseptic-processing>)

Required Level: Master student with experience in computer vision, robotics and deep learning. You should have performed a deep learning project before.

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